

Membrane Air Separation

Final Lab Report
Unit Operations Lab 2
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1. Executive Summary (Examples at end of the Guideline) (10 pts)

The objective is to characterize the pressure-driven permeation of O_2/N_2 by using measured flow rates and oxygen percentages from a hollow fiber membrane air separation rig. Two configurations, parallel and series, are compared in terms of performance across transport coefficients and separation factors. Performances are examined by plotting these as functions of operating pressure, with tube-side gas flow rate and parallel and series as parameters. Following this analysis, conclusions are drawn from the results with respect to pressure, rate, and configuration. Finally, observations on the correlation between diffusivity, separation, and recovery. Also, determine the correlation between O_2 and N_2 diffusivity differences.

The overall mass transfer coefficient, separation factors, and percent recoveries were obtained for the air gas mixture consisting of O_2 and N_2 . Moreover, these values were found for the parallel and series configurations of the apparatus. They were found at 3 different pressure intervals at varying flow rates in each interval.

Based on the data collected and the calculations completed, the correlation between flow rate and pressure for diffusivity, overall mass transfer coefficients, and recoveries were found. The correlation was successful in determining the optimum separation conditions and O_2 V N_2 diffusivity coefficients. Also, the correlation successfully displayed the correlation between separation, flow rate, and pressure. Finally, the diffusivity of O_2 and N_2 aligns with the separation and recovery of O_2 and N_2 .

2. Introduction (10 pts)

Synthetic or modified natural polymeric membrane-based processes are commonly used in a wide range of commercial technologies. Chemical businesses use membranes as a separation technique, such as component gas separation and desalination. Nonporous membrane treatment of gas streams makes it possible to recover component gases without resorting to more expensive methods like liquefaction or evaporation. In the separation process known as membrane permeation, gas molecules are selectively transported via a permeable polymeric screen. This type of membrane separation is primarily dependent on the relative rates of mass transfer, as opposed to most chemical engineering separation processes, which are controlled by phase equilibrium relations.

Adsorption of the gas molecule onto the interface of the high-pressure side of the membrane, dissolving of the gas into the membrane at the interface, and diffusion of the gas through the membrane comprise the solution-diffusion mechanism that will cause transport. Desorption of the gas molecule from the membrane's low-pressure side · elution of the gas from the membrane at the contact.

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Experimental objective, significance of the experiment, brief description of the apparatus and experiments, and **industrial applications**.

3. Theory (10 pts)

A membrane is a physical barrier between 2 fluids (the feed and product sides) that selectively allows certain components of the feed fluid to pass. The fluid that passes through the membrane is referred to as permeate, while retentate is the term for the fluid that is kept on the feeding side[1]. The tools required for separation are quite basic. The membrane and a container containing the two fluids make up this structure. A stirred tank divided into two volumes by a membrane is the most basic configuration. Although they are not frequently utilized in large-scale separations, batch and continuous stirred-tank systems are utilized in lab settings[1].

A mix of diffusion and turbulence transports fluid to the membrane wall. The driving force, which is often a pressure differential across the membrane, pushes through the membrane at the membrane wall[1]. Diffusion and turbulent mixing sweep the material that was transported toward the wall but rejected by the membrane away from the membrane surface. It should be noted that we do not want to deposit this material on the membrane wall, in contrast to standard filtration, which forms filter cakes[1]. Rather, the concentration of the permeate decreases while that of the retentate increases.

The plate-and-frame system is comparable to a plate-and-frame filter press or a parallel-plate heat exchanger, with the exception that flat membrane sheets are used in place of the filter cloth[1]. This design is utilized for electrodialysis (which includes running a current across the membrane), for food processing applications where thorough cleaning by disassembly may be necessary, and for membrane materials that are challenging to form into more complex shapes[1].

The parallel configuration facilitates an increase in the feed flow rate and is applicable to all types of membrane separations[1]. Operating in parallel can be compared to enhancing the diameter of a distillation or absorption column, although there is minimal economy of scale. The retentate-in-series system is typically employed when the retentate serves as the product[1]. This cascade method enhances the purity of the more strongly retained components while also boosting the recovery of the more permeable species. However, the purity of the permeate and the recovery of retained components both tend to diminish[1]. This setup is utilized for producing less-permeable nitrogen from air's oxygen at relatively high pressure, as nitrogen passes through the membrane at a significantly slower rate than oxygen[1]. This system can be likened to a crossflow extractor or stripper. Often, parallel and retentate-in-series systems are integrated.

The permeate-in-series system is implemented when the purity of the permeate product is inadequate, necessitating one or more additional separation stages. Designers

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prefer to avoid this arrangement when possible[1]. The main operational cost of most membrane separators stems from the energy needed to establish the pressure difference that drives permeate through the membrane. This configuration demands an extra compressor or pump to re-pressurize the permeate for the subsequent membrane separator[1]. A significant drawback of membrane separators is that additional separation stages for the permeate do not utilize the energy-separating agent (pressure) again. In contrast, equilibrium-staged separations benefit from the ability to easily recycle energy-separating agents (heat and cooling). Ideally, a commercial membrane would exist that yields permeate of the required purity in a single stage[1].

The patterns of flow on both the retentate (feed) and permeate sides of the membrane significantly influence the mass balances and the separation process[1]. These flow patterns can include complete mixing on both sides, plug flow on one side with complete mixing on the other, cocurrent plug flow on both sides, countercurrent plug flow on the two sides, mixed flow on one side combined with crossflow on the other, or variations between these ideal scenarios[1]. For instance, when analyzing the hollow-fiber module, the flow within the hollow fibers closely resembles plug flow. Depending on the design of the shell side, the flow may be approximately well-mixed, exhibit cocurrent plug flow, or feature countercurrent plug flow. Typically, countercurrent plug flow results in the best separation outcomes[1].

Undoubtedly, the critical element of membrane separators is the membrane itself[1]. The membrane should exhibit high permeability for the components intended to become the permeate product while demonstrating low permeability for the components meant to be retained in the retentate. It is beneficial for the membrane to have high resistance to temperature and chemicals, be mechanically robust, minimize fouling, be capable of being shaped into the required module configurations, and remain cost-effective[1]. Commercial membranes are typically composed of polymers, ceramics, and metals (such as palladium for helium purification), but polymer membranes are predominantly utilized[1]. The development of membranes is carried out by a handful of large chemical corporations and various niche firms. While many chemical engineers utilize membrane separators, they often do not participate in membrane development. Nonetheless, a basic understanding of polymer membranes is advantageous when specifying and managing membrane separators[1].

A wide array of polymers has been utilized to manufacture membranes for separation processes[1]. The initial commercially successful membranes were made from cellulose, ethyl cellulose, and cellulose acetate (which is a polymer blend comprising cellulose, cellulose acetate, and cellulose triacetate); these materials are still in use today[1]. Although these substances exhibit relatively poor chemical resistance, their low cost makes them appealing for appropriate applications. Polysulfone is the most used commercial membrane material due to its excellent chemical and thermal resistance. Various polyamide polymers are frequently employed in reverse osmosis (RO)[1]. Silicone rubber, which offers very high flux rates but lacks selectivity, is commonly used in pervaporation to preferentially allow organic compounds to permeate and acts as a

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coating on composite membranes for gas permeation (GP)[1]. A diverse range of grafted polymers, specialized polymers, and composite membranes has been created to enhance flux and selectivity, particularly for pervaporation as well as for other membrane separation processes[1].

Permeability, Q , is a transport coefficient that can be measured directly through experimental methods[1]. In certain instances, permeability can be approximated from more basic variables like solubility and diffusivity[1]. The separation thickness of the membrane refers to the thickness of the part of the membrane that is actively engaged in the separation process. And the flux is the flow rate per unit area, increasing flux decreases membrane area. The result is a more compact, less expensive device[1].

$$\alpha_{AB} = \frac{Q_A}{Q_B} \quad [\text{Separation efficiency}]$$

$$\alpha'_{AB} = \frac{\frac{X_{Ap}}{X_{Bp}}}{\frac{X_{Ar}}{X_{Br}}} \quad [\text{Separation factor based on retentate composition}]$$

$$\alpha''_{AB} = \frac{\frac{X_{Ap}}{X_{Bp}}}{\frac{X_{Af}}{X_{Bf}}} \quad [\text{Separation factor based on feed composition}]$$

$$J_A = \frac{q_A}{z} (p_{A1} - p_{A2}) = Q_A (p_{A1} - p_{A2})$$

$$Q_A = \text{Permeability} \left[\frac{\text{mol}}{\text{smPa}} \right]$$

p_{A1} = Partial Pressure of component A on feed side (tube)

p_{A2} = Partial Pressure of component A on the permeate side (shell)

$$F = K_A (p_{A1} - p_{A2})$$

$$F = \text{Species A flowrate} \left[\frac{\text{mol}}{\text{min}} \right]$$

$$K_A = \text{Overall Mass Transfer Coefficient} \left[\frac{\text{mol}}{\text{min Pa}} \right]$$

$$O_2 \text{ Recovery} = \frac{V_p * C_{O2p}}{V_f * C_{O2f}}$$

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$$N_2 \text{ Recovery} = \frac{V_r * C_{N_2p}}{V_f * C_{N_2f}}$$

$$STAGECUT = \frac{Q_p}{Q_p + Q_r}$$

4. Results (15 pts)

The overall mass transfer coefficients, KO₂ and KN₂, were calculated from the measured permeate molar flows and the tube side mass flow rate. KO₂ is the transport coefficient for oxygen, while KN₂ represents the same parameter for nitrogen. The ratio of KO₂/ KN₂ serves as an experimental measure of membrane selectivity, with larger values indicating the preferential selectivity of the membrane to oxygen. In table 1 and table 2 there is a general trend of an increasing KO₂/KN₂ ratio as the mass flow rate increases. Comparing across operating pressures the KO₂/NO₂ ratio decreases as pressure increases.

Table 1: Comparing KN₂ and KO₂ Values parallel

mass flow (gmol/min)	KO₂	KN₂	Ratio KO₂/KN₂
51 PSIG			
0.04375	0.000152763	4.32734E-05	3.530
0.094196429	0.00017238	4.3684E-05	3.946
0.142410714	0.000187799	4.37982E-05	4.288
0.184821429	0.000199605	4.39874E-05	4.538
0.227678571	0.000206243	4.49311E-05	4.590
120 PSIG			
0.070982143	0.000169768	5.0647E-05	3.352
0.091964286	0.00017744	5.07282E-05	3.498
0.133482143	0.000186697	5.11491E-05	3.650
0.178571429	0.000196951	5.11525E-05	3.850
0.223214286	0.000205421	5.12286E-05	4.010
180 PSIG			
0.048660714	0.000172105	5.51319E-05	3.122
0.089285714	0.000179624	5.49113E-05	3.271
0.133928571	0.000186426	5.47557E-05	3.405
0.177232143	0.000194701	5.4392E-05	3.580
0.223214286	0.000201726	5.42403E-05	3.719

Figure 1: Scatter plot of relationship between KO_2 and mass flow rate (in parallel)

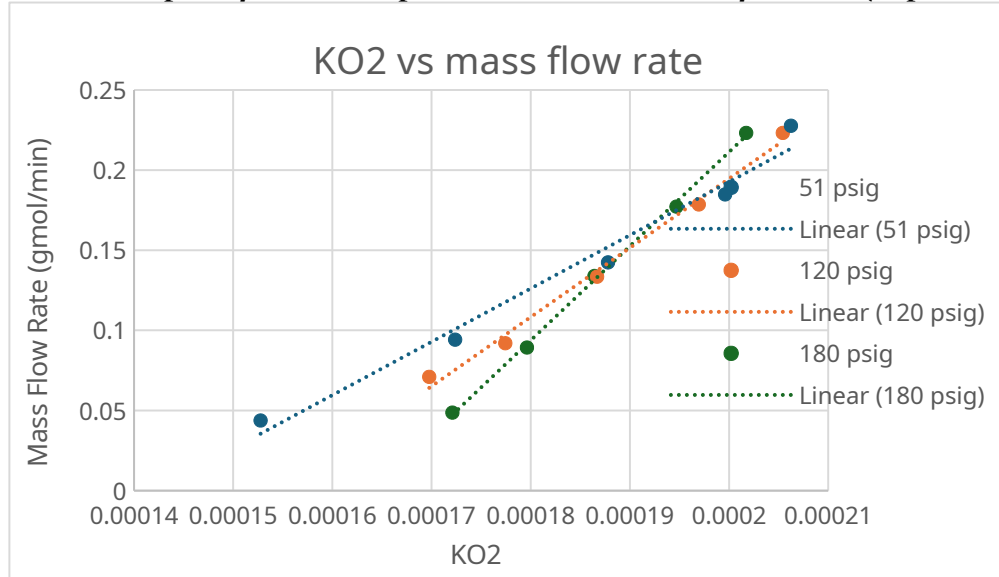


Figure 2: Scatter plot of relationship between KN_2 and mass flow rate (In parallel)

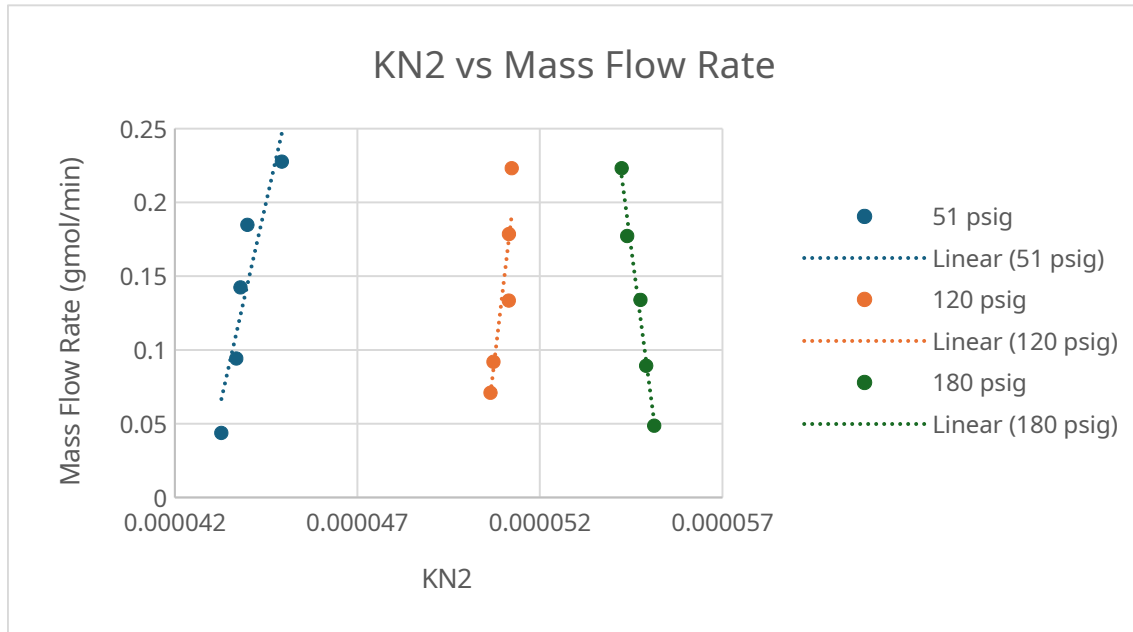


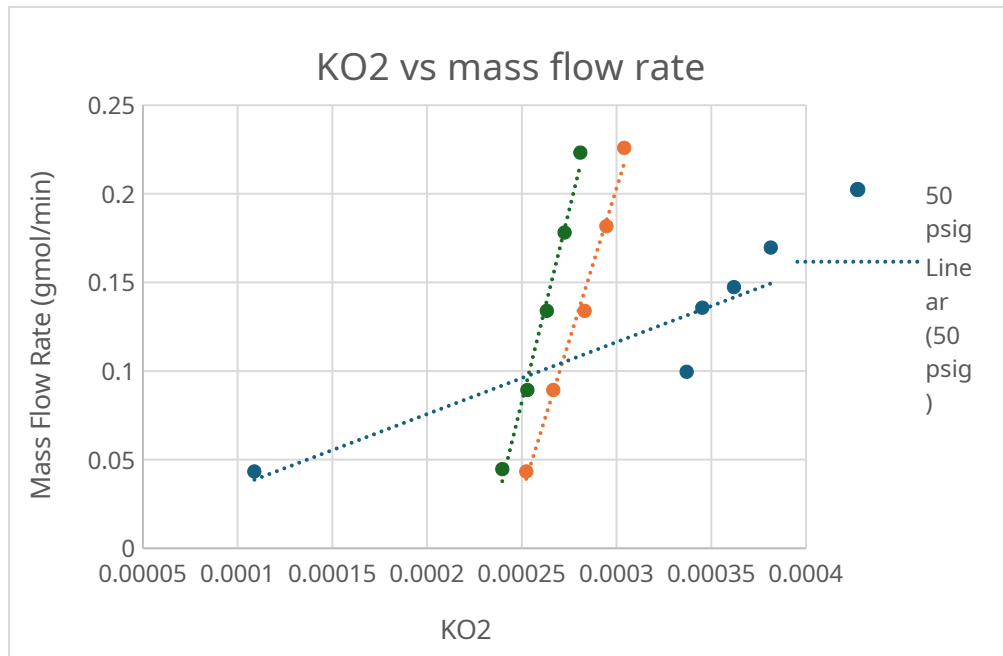
Table 2: Comparing KN2 and KO2 Values series

Flow Tube Pipe (gmol/min)	KO2	KN2	Ratio KO2/KN2
50 PSIG			
0.043303571	0.000108841	5.60203E-05	1.943
0.099553571	0.000337001	4.21545E-05	7.994
0.135714286	0.000345279	4.20534E-05	8.210
0.147321429	0.000361957	4.12766E-05	8.769
0.169642857	0.000381411	4.17197E-05	9.142
120 PSIG			
0.043303571	0.000252379	4.87635E-05	5.176
0.089285714	0.000266616	4.86193E-05	5.484
0.133928571	0.00028317	4.77122E-05	5.935
0.181696429	0.000294659	4.7183E-05	6.245
0.225892857	0.000304064	4.65633E-05	6.530
180 PSIG			
0.044642857	0.000239728	5.11763E-05	4.684
0.089285714	0.00025296	5.03573E-05	5.023

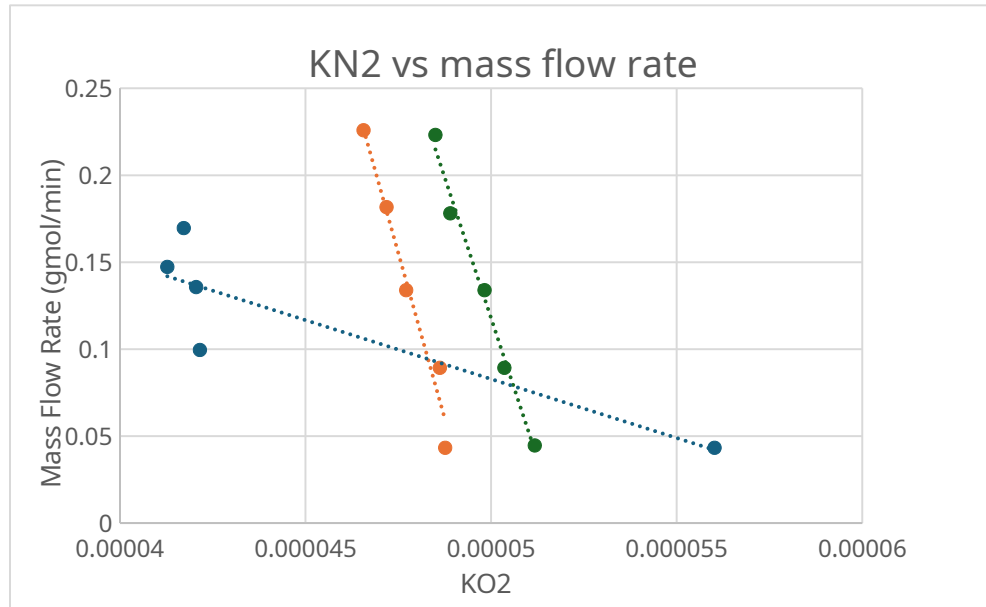
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0.133928571	0.000263193	4.9823E-05	5.283
0.178125	0.000272581	4.8901E-05	5.574
0.223214286	0.000280927	4.84996E-05	5.792

Figure 4: Scatter plot of relationship between KO₂ and mass flow rate (in series)



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Alpha is the value that determines the separation efficiency of O₂ and N₂. Alpha prime is the separation factor based on the retentate composition. Alpha double prime is the separation actor based on the feed composition.

Table 3: Alpha values (in Parallel)

alpha	alpha'	alpha''
51 PSIG		
3.530	4.240	1.758
3.946	4.155	1.763
4.288	4.098	1.768
4.538	4.072	1.773
4.590	3.970	1.764
120 PSIG		
3.352	7.712	1.995
3.498	7.850	2.009
3.650	7.645	2.011
3.850	7.445	2.016
4.010	7.312	2.021
120 PSIG		
3.122	10.311	2.067
3.271	10.016	2.072
3.405	9.729	2.075
3.580	9.762	2.089
3.719	9.542	2.094

Table 4: Alpha Values (in series)

alpha	alpha'	alpha''
50 PSIG		
1.943	3.457	1.623
7.994	6.739	2.051
8.210	6.442	2.043
8.769	6.245	2.049
9.142	6.061	2.050
120 PSIG		
5.176	16.955	2.280
5.484	17.204	2.301
5.935	14.975	2.307
6.245	13.798	2.312
6.530	12.878	2.317
180 PSIG		
4.684	22.495	2.321
5.023	23.832	2.354
5.283	22.129	2.365
5.574	20.262	2.376
5.792	19.178	2.385

This section will go over the recovery separation performance of oxygen and nitrogen across the membrane system at varying pressures. Oxygen recovery and nitrogen recovery can be determined by using measured volumetric flow rates and gas compositions. These results were derived from experimental data by treating air as an ideal gas and relating mole fractions to molar concentrations. The following figures and tables go over the calculations and illustrate how membrane behavior changes with operating pressure. They also show the behavioral changes between parallel membrane separation and series membrane separation.

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Table 3: Oxygen Recovery at Varying Feed Pressures in Parallel

Vp (m³/min)	Vf (m³/min)	Vr (m³/min)	O2 Recovery %
51 PSIG			
0.003250650			44.36
0	0.004230846	0.003537996	
0.003400680			38.62
0	0.005511102	0.007617522	
0.003510702			34.82
0	0.00670134	0.011516538	
0.003600720			32.17
0	0.007741548	0.014946228	
0.003710742			29.7
0	0.008811762	0.01841202	
120 PSIG			
0.008861772			16.59
0	0.01045209	0.005740218	
0.008981796			16.35
0	0.011042208	0.007437012	
0.009191838			15.72
0	0.012182436	0.010794498	
0.009371874			15.26
0	0.013372674	0.0144408	
0.009531906			14.76
0	0.014532906	0.018051	
180 PSIG			
0.014172834			10.98
0	0.015263052	0.003935118	
0.014322864			10.78
0	0.016323264	0.0072204	
0.014472894			10.53
0	0.017473494	0.0108306	
0.014602920			10.37
0	0.018573714	0.014332494	
0.014762952			10.17
0	0.019763952	0.018051	

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Referring to Table 1, the data can be characterized based on the operating pressures; as the overall system pressure increases, the O₂ recovery decreases. When the operating pressure is held constant, we observe that recovery is highest at lower volumetric flow rates (V_p and V_r). This behavior is expected because a lower flow rate increases the contact time of the gas within the membrane module, allowing more oxygen to permeate through the membrane. The extended contact time enhances the driving force utilization across the membrane, resulting in higher oxygen recovery. These results indicate that membrane efficiency is dependent on flow rate and operating pressure, where oxygen recovery improves as flow rate increase and pressure decrease.

Table 4: Nitrogen Recovery at Varying Feed Pressures in parallel

Volumetric Flow of Permeate (m ³ /min)	Vf (m ³ /min)	Vr (m ³ /min)	N2 recovery (%)
51 PSIG			
0.0032506500	0.004230846	0.003537996	5.96
0.0034006800	0.005511102	0.007617522	9.72
0.0035107020	0.00670134	0.011516538	11.95
0.0036007200	0.007741548	0.014946228	13.31
0.0037107420	0.008811762	0.01841202	14.31
120 PSIG			
0.0088617720	0.01045209	0.005740218	1.99
0.0089817960	0.011042208	0.007437012	2.44
0.0091918380	0.012182436	0.010794498	3.20
0.0093718740	0.013372674	0.0144408	3.88
0.0095319060	0.014532906	0.018051	4.44
180 PSIG			
0.0141728340	0.015263052	0.003935118	0.66
0.0143228640	0.016323264	0.0072204	1.12
0.0144728940	0.017473494	0.0108306	1.57
0.0146029200	0.018573714	0.014332494	1.95

0.0147629520 0.019763952 0.018051 **2.30**

We observe that nitrogen recovery follows the inverse trend of oxygen recovery. Whereas oxygen recovery is highest at lower flow rates, nitrogen recovery is highest when both the retentate volumetric flow rate (V_r) and the permeate flow rate (V_p) are relatively large at the same operating pressure (Table 2). This behavior is expected because nitrogen recovery reflects the fraction of nitrogen that remains in the retentate stream. At higher flow rates, the gas spends less time in contact with the membrane, reducing nitrogen permeation into the shell and therefore increasing the fraction of nitrogen retained. This is the opposite of oxygen recovery, which generally decreases with increasing flow because there is less time for oxygen to permeate. When comparing recovery at different operating pressures, nitrogen recovery is still lower at higher pressures, consistent with the increased driving force that promotes additional nitrogen permeation.

Table 5: Oxygen Recovery at varying Feed pressures in series

V_f (m ³ /s)	V_p 1(m ³ /s)	V_r 2 (m ³ /s)	O2 recovery (%)
50 PSIG			
0.000275579	0.000217214	5.83649E-05	32.82
0.000359215	0.000225036	0.000134179	49.55
0.000411563	0.000228646	0.000182917	45.27
0.000430216	0.000231655	0.000198561	45.92
0.000468724	0.000240078	0.000228646	45.21
120 PSIG			
0.00062276	0.000564395	5.83649E-05	20.82
0.000693158	0.000572818	0.00012034	19.60
0.000762354	0.000581844	0.00018051	19.27
0.000834558	0.000589666	0.000244892	18.59
0.000900143	0.000595683	0.00030446	18.08
180 PSIG			
0.000932635	0.000872465	0.00006017	13.37
0.000997017	0.000876677	0.00012034	13.06
0.001065009	0.000884499	0.00018051	12.79

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0.001128188	0.000888109	0.000240078	12.63
0.001196781	0.000895931	0.00030085	12.36

As the overall system pressure increases, the recovery of O₂ also decreases (Table 3). When keeping the pressure constant, with increasing permeate volumetric flow rate (V_p) and increasing feed flowrate (V_f) the O₂ recovery increases. This trend suggests that to get optimal O₂ recovery in a series configuration we need to have a low pressure as well low flow rates so that the air has maximized contact time with the membrane. This indicates that membrane efficiency is dependent on operating pressure as well as flow rate.

Table 6: Nitrogen Recovery at varying Feed pressures in series

V _f (m ³ /s)	V _p 1(m ³ /s)	V _r 2 (m ³ /s)	N ₂ Recovery (%)
50 PSIG			
0.000275579	0.000217214	5.83649E-05	19.18
		0.00013417	
0.000359215	0.000225036	9	29.77
		0.00018291	
0.000411563	0.000228646	7	34.96
		0.00019856	
0.000430216	0.000231655	1	35.60
		0.00022864	
0.000468724	0.000240078	6	37.01
120 PSIG			
0.00062276	0.000564395	5.83649E-05	2.70
0.000693158	0.000572818	0.00012034	4.95
0.000762354	0.000581844	0.00018051	6.62

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		0.00024489	
0.000834558	0.000589666	2	8.08
0.000900143	0.000595683	0.00030446	9.19
180PSIG			
0.000932635	0.000872465	0.00006017	1.19
0.000997017	0.000876677	0.00012034	2.21
0.001065009	0.000884499	0.00018051	3.07
		0.00024007	
0.001128188	0.000888109	8	3.81
0.001196781	0.000895931	0.00030085	4.47

Referring to table 4 we observed that the recovery of N₂ follows a inverse trend to that of oxygen recovery. Where Oxygen recovery was the most optimal at lower pressures and flowrates compared to that of N₂ recovery we see it's the most optimal at higher V_f and V_p for whichever operating pressure. This behavior is to be expected as the higher the flow rates the less time the air has in contact with the membrane reducing nitrogen permutation therefore increasing the fraction of nitrogen retained. When comparing recovery at different operating pressures, nitrogen recovery is still lower at higher pressures, consistent with the increased driving force that promotes additional nitrogen permeation.

5. Discussion (15 pts)

In this lab, the configuration of parallel and series for air membrane separation was conducted. The configurations done are to be compared for optimal flow rates and pressure combinations. Moreover, the difference between series and parallel configurations lies in the difference in flow rate and pressure tolerance in the system. While parallel allows a higher flow rate and pressure. A series doesn't have the same tolerance to flow rate and pressure, which can allow better separation and performance.

As for the comparison between the results and the theory of the lab a clear agreement, as O₂ has a much better permeation and separation than N₂. For example, in

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parallel, the value of KO₂ is bigger by 2 decimal places when compared to KN₂. For reference in the Unit Operations 5th edition, the diffusion coefficients in natural rubber are 1,580 for O₂ and 1,110 for N₂. There is a distinct reason for the large difference; the molecular size and molecular shape of O₂ are favored in polysulfone fibers [2]. In literature, the diffusion coefficient is directly linked to the “free volume” of the polymer [2]. For glassy polymers like polysulfone, the process of diffusion depends on the size of the molecule, in our case, O₂ and N₂ [2]. The molecular size of N₂ is slightly larger than that of O₂, which allows O₂ to have a better diffusivity than N₂.

In series, we see the same correlation of diffusion with KO₂ trumping KN₂. The difference in numbers between O₂ and N₂ is in their respective percentage recoveries. In literature, series configuration allows for higher recovery in the more permeable species in the membrane [3]. This is shown when comparing Table 1 and Table 3 in the results section. There is a clear increase in recovery for O₂ in series configuration when compared to parallel. This agrees with what the theory of the literature states, as well as what has been stated for parallel configuration.

When discussing the separation factor and efficiency, α , it's a ratio of O₂/N₂ separation. What this says, according to literature, is how well the membrane separation system separated the desired components from a mixture [4]. As previously discussed, O₂ has a better diffusion so from the results of α we can see in Tables 3 and 4 that the ratio is a larger number, indicating that O₂ has a better separation efficiency when compared to N₂. A higher separation efficiency also indicates a more effective and robust process, allowing for higher purity products and recovery of the desired product [4]. The α in a series configuration is much higher than in parallel, which can be supported by the literature. In the literature, a series of repeated filtration and an increased driving force allows for better purity and separation than parallel [4].

Next, the value of the separation factor is based on the retentate composition, α'_{AB} . According to the literature α'_{AB} is a dimensionless quantity that measures how effectively a separation process can differentiate between 2 components in a mixture [5]. Also, according to the same literature, when $\alpha = 1$ no separation has occurred as the components have no differentiation in the membrane. Moreover, when $\alpha > 1$ the process successfully separated the components in the mixture [5]. The same correlation is applied to α''_{AB} . As it is the same metric, just a ratio between feed and not retentate. The α'_{AB} shows to have the highest separation factor for the reason that the retentate feed is smaller, allowing for the separation ratio to be bigger in series and parallel.

As for the percentage recoveries for O₂ and N₂, these can be related back to the theory for the KO₂ and KN₂ discussion. The recovery of O₂ is always higher than N₂, as shown in tables 3, 4, 5, and 6. The reason for O₂ having more recovery than N₂ can be referred back to the size difference and the diffusivity difference between O₂ and N₂. With O₂ having the diffusivity advantage, the membrane can be more selective towards O₂ in both series and parallel configurations. The relationship between the overall mass

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transfer coefficient, K , and the percent recovery is linearly correlated, hence the similar explanation and theory. Additionally, the recovery of O_2 is higher in series, aligning with the nature and theory of series configuration, allowing better purity and recovery output in the membrane.

Possible areas of error can lie within the apparatus, such as not purging the streams enough when changing flow rates and pressure. Also, the O_2 sensors are not reading or calibrating correctly. A pressure leak from the compressed air tank is causing a loss of pressure in the system. Moreover, the flow meter can be faulty and not allow the correct amount of flow into the system. The flowmeter can also be mis-calibrated with the controller and the valve not matching the sensitivity of the meter. With all these causes of error, none of these affected the ability to yield good results. These errors are only hypothetical and don't apply to the experiment completed. Preventative measures for these causes of error are checking the pressure hose going into the system, calibrating the O_2 sensor before the experiment, and allowing 10 minutes of calibration time. Also, make sure the flow meter matches the controller by calibrating. Finally, purging the valves when changing the flow rates and pressure going into the system.

For the discussion of the trend between flow rate, pressure, and configuration based on the results, higher pressure increases permeation flux, but if the pressure reaches a high threshold, it reduces selectivity as it's pushing the gas mixture and forcing it through the membrane. Per pressure interval, the lower flow rates allow more residence time for separation to occur more precisely. If the flow rates are too high, similar to the pressure, the sweeping gas in the system has less contact with the membrane and does not allow separation to occur. As for parallel and series configuration, series allows for better separation due to more filtration and stages required to separate the mixture.

For optimal parameters for each alpha, it can be broken down by each type. For α in series, the optimal separation is 9.142. The parameters that allow this are a flow rate of 3.99 SLPM and a pressure of 50 PSIG. Next α'_{AB} in series, the optimal value is 23.832, and the parameters are a flow rate of 14.57 SLPM and a pressure of 180 PSIG. Finally, α''_{AB} in series with an optimal value of 2.385 with a flow rate of 14.89 SLPM and a pressure of 180 PSIG.

Now, for optimal parameters in parallel for each alpha variable. For α , an optimal value is 4.590 SLPM with a flow rate of 3.71 and a pressure of 51 PSIG. For α'_{AB} , an optimal value of 10.311 and a flow rate of 14.17 SLPM and a pressure of 180 PSIG. Finally, α''_{AB} , an optimal value of 2.094 with a flow rate of 14.76 SLPM and a pressure of 180 PSIG.

The correlation between the separation efficiency and factor in parallel and series is that α in series and parallel are most optimal at low flow rate and low pressure. For α'_{AB} and α''_{AB} the correlation is high pressure and high flow rate.

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All the molar concentrations were calculated under the assumption that the air used behaves ideally under the experimental conditions. This implies that the intermolecular interactions are negligible as the gas molecules occupy no volume relative to the container. This allows the use of the ideal gas law, where $C = P/RT$ is utilized to define the molar concentrations needed to calculate the oxygen and nitrogen recovery rates. This assumption is generally reliable for experiments operating at relatively low pressures and near-ambient temperatures. Under these conditions, air behaves like an ideal gas where the compressibility factor can be taken as equal to 1.

6. References (10 pts)

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7. Appendix I: Data Tabulation/Graphs (10 pts)

Run #	Pressure	Shell side	mass flow shell	o2 top (%)
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	(PSIG)	Pipe(SLPM)	(gmol/min)	
1	51	3.25	0.145	29.8
2	51	3.4	0.151	32.3
3	51	3.51	0.156	34.3
4	51	3.6	0.161	35.7
5	51	3.71	0.166	36.4
6	120	8.86	0.396	29.3
7	120	8.98	0.401	30.1
8	120	9.19	0.410	31.2
9	120	9.37	0.418	32.6
10	120	9.53	0.425	33.7
11	180	14.17	0.633	27.8
12	180	14.32	0.639	28.9
13	180	14.47	0.646	29.9
14	180	14.6	0.652	31
15	180	14.76	0.69	32

x1s	Flowrate Bottom Pipe(SLPM)	mass flow bot(gmol/min)	o2 bot (%)	x1b
0.298	0.98	0.043	9.1	0.091
0.323	2.11	0.094	10.3	0.103
0.343	3.19	0.142	11.3	0.113
0.357	4.14	0.185	12	0.12
0.364	5.1	0.227	12.6	0.126
0.293	1.59	0.071	5.1	0.051
0.301	2.06	0.091	5.2	0.052
0.312	2.99	0.133	5.6	0.056
0.326	4	0.179	6.1	0.061
0.337	5	0.223	6.5	0.065
0.278	1.09	0.041	3.6	0.036
0.289	2	0.089	3.9	0.039
0.299	3	0.133	4.2	0.042
0.31	3.97	0.177	4.4	0.044
0.32	5	0.223	4.7	0.047

Inlet Flowrate (SLPM)	Mass Flow	Pressure (Torr)	Vp (m ³ /s)	Vf (m ³ /s)	Vp (mol/m ³)	mol/ m ³ Vb
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					3)	
			0.00005	0.00007		
4.23	0.19	3396.73	4	1	0.11	3.66
			0.00005	0.00009		
5.51	0.25	3396.73	7	2	0.12	4.14
			0.00005	0.00011		
6.70	0.30	3396.73	9	2	0.13	4.54
			0.00006	0.00012		
7.74	0.35	3396.73	0	9	0.13	4.82
			0.00006	0.00014		
8.81	0.39	3396.73	2	7	0.13	5.06
			0.00014	0.00017		
10.45	0.47	6964.08	8	5	0.11	2.05
			0.00015	0.00018		
11.04	0.49	6964.08	0	4	0.11	2.09
			0.00015	0.00020		
12.18	0.54	6964.08	3	3	0.11	2.26
			0.00015	0.00022		
13.37	0.60	6964.08	6	3	0.12	2.45
			0.00015	0.00024		
14.53	0.65	6964.08	9	3	0.12	2.61
			0.00023	0.00025		
15.26	0.68	10066.12	7	5	0.10	1.44
			0.00023	0.00027		
16.32	0.73	10066.12	9	3	0.11	1.56
			0.00024	0.00029		
17.47	0.78	10066.12	2	2	0.11	1.69
			0.00024	0.00031		
18.57	0.83	10066.12	4	0	0.11	1.77
			0.00024	0.00033		
19.76	0.88	10066.12	6	0	0.1	1.89

O2 (in)	O2 (out)	P1A	P1b	p1t
0.04	0.04	709.91	309.11	509.51
0.05	0.05	709.91	349.86	529.89
0.06	0.07	709.91	383.83	546.87
0.07	0.08	709.91	407.61	558.76

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0.08	0.08	709.91	427.99	568.95
0.10	0.12	1455.49	355.17	905.33
0.10	0.13	1455.49	362.13	908.81
0.11	0.14	1455.49	389.99	922.74
0.12	0.15	1455.49	424.81	940.15
0.14	0.16	1455.49	452.67	954.07
0.14	0.18	2103.81	362.38	1233.1
0.15	0.18	2103.81	392.58	1248.19
0.16	0.19	2103.81	422.78	1263.29
0.17	0.21	2103.81	442.91	1273.36
0.18	0.22	2103.81	473.11	1288.46

p1s	g1
226.48	0.04
245.48	0.04
260.68	0.05
271.32	0.05
276.64	0.06
222.68	0.12
228.76	0.12
237.12	0.12
247.76	0.14
256.12	0.14
211.28	0.18
219.64	0.18
227.24	0.19
235.6	0.20
243.2	0.21

x2r	x2a	x2s	x2b	P2A
0.21	0.7910	0.70	0.91	2686.81
0.22		0.68	0.89	2686.81
0.23		0.66	0.89	2686.81
0.24		0.64	0.88	2686.81
0.23		0.64	0.87	2686.81
0.24		0.71	0.95	5508.58

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0.25	0.70	0.95	5508.58
0.26	0.69	0.94	5508.58
0.27	0.67	0.94	5508.58
0.27	0.66	0.94	5508.58
0.24	0.72	0.96	7962.30
0.25	0.71	0.96	7962.30
0.26	0.70	0.96	7962.30
0.27	0.69	0.96	7962.30
0.27	0.68	0.95	7962.30

P2b	p2t	p2s	g2
3087.63	2887.22	533.52	0.10
3046.87	2866.84	514.52	0.10
3012.90	2849.86	499.32	0.10
2989.13	2837.97	488.68	0.10
2968.74	2827.78	483.36	0.11
6608.91	6058.75	537.32	0.28
6601.95	6055.27	531.24	0.28
6574.09	6041.34	522.88	0.28
6539.27	6023.93	512.24	0.28
6511.41	6010.00	503.88	0.28
9703.74	8833.02	548.72	0.46
9673.54	8817.92	540.36	0.46
9643.34	8802.82	532.76	0.45
9623.21	8792.76	524.40	0.45
9593.01	8777.66	516.80	0.45

alpha	alpha'	alpha''
3.53	4.24	1.76
3.95	4.15	1.76
4.28	4.09	1.77
4.54	4.07	1.77

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4.59	3.97	1.76
3.35	7.71	1.99
3.50	7.85	2.00
3.65	7.64	2.01
3.85	7.45	2.01
4.01	7.31	2.03
3.12	10.31	2.07
3.27	10.01	2.07
3.40	9.72	2.08
3.58	9.76	2.09
3.72	9.54	2.09

Vp (m ³ /min)	Vf (m ³ /min)	Vr (m ³ /min)	O2 recovery
0.0033	0.0042	0.0010	0.4436
0.0034	0.0055	0.0021	0.3862
0.0035	0.0067	0.0032	0.3482
0.0036	0.0077	0.0041	0.3217
0.0037	0.0088	0.0051	0.2970
0.0089	0.0105	0.0016	0.1659
0.0090	0.011	0.0021	0.1635
0.0092	0.0122	0.0030	0.1572
0.0094	0.0134	0.0040	0.1526
0.0095	0.0145	0.0050	0.1476
0.0142	0.0153	0.0011	0.1098
0.0143	0.0163	0.0020	0.1079
0.0145	0.0175	0.0030	0.1054
0.0146	0.0186	0.0040	0.1037
0.0148	0.0198	0.0050	0.1017

O2 Recovery (%)	N2 recovery	N2 recovery (%)
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44.36	0.059	5.950
38.62	0.097	9.713
34.82	0.119	11.948
32.17	0.133	13.313
29.71	0.143	14.314
16.59	0.019	1.992
16.35	0.024	2.440
15.72	0.032	3.197
15.26	0.038	3.875
14.76	0.044	4.439
10.98	0.006	0.657
10.79	0.011	1.124
10.54	0.015	1.572
10.37	0.019	1.951
10.17	0.023	2.307

Flow Tube Pipe (gmol/min)	o2 bot (%)	x1b	Inlet Flowrate (SLPM)	flow tube pipe (gmol/min)
0.04	7.10	0.07	4.58	0.20
0.10	8.90	0.09	5.97	0.27
0.14	9.70	0.10	6.84	0.31
0.15	10.70	0.11	7.15	0.32
0.17	11.60	0.12	7.79	0.35
0.04	3.00	0.03	10.35	0.46
0.09	3.10	0.03	11.52	0.51
0.13	3.90	0.04	12.67	0.57
0.18	4.50	0.05	13.87	0.62
0.23	5.10	0.05	14.96	0.67
0.04	2.20	0.02	15.50	0.69
0.09	2.20	0.02	16.57	0.74
0.13	2.50	0.03	17.70	0.79
0.18	2.90	0.03	18.75	0.84
0.22	3.20	0.03	19.89	0.89

Ptotal	p1a	p1b	p1s	p1t
3345.03	699.11	237.50	158.84	468.30
3345.03	699.11	297.71	301.72	498.41
3345.03	699.11	324.47	310.84	511.79
3345.03	699.11	357.92	325.28	528.52
3345.03	699.11	388.02	336.68	543.57
6964.08	1455.49	208.92	261.44	832.21
6964.08	1455.49	215.89	269.80	835.69
6964.08	1455.49	271.60	287.28	863.55
6964.08	1455.49	313.38	299.44	884.44
6964.08	1455.49	355.17	310.84	905.33
10066.12	2103.82	221.45	255.36	1162.64
10066.12	2103.82	221.45	265.24	1162.64
10066.12	2103.82	251.65	275.12	1177.74
10066.12	2103.82	291.92	286.52	1197.87
10066.12	2103.82	322.12	294.88	1212.97

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g1	K02	retentate		
		flow	xAf	x1f
0.03	0.00011	3.37	0.06	0.14
0.07	0.00034	6.62	0.13	0.24
0.07	0.00035	6.93	0.15	0.25
0.07	0.00036	7.34	0.17	0.27
0.08	0.00038	7.87	0.19	0.28
0.14	0.00025	14.40	0.17	0.19
0.15	0.00027	15.08	0.20	0.19
0.16	0.00028	16.31	0.24	0.21
0.17	0.00029	17.23	0.27	0.22
0.18	0.00030	18.06	0.31	0.23
0.22	0.00024	21.75	0.25	0.18
0.23	0.00025	22.70	0.27	0.19
0.24	0.00026	23.75	0.31	0.19
0.25	0.00027	24.84	0.34	0.20
0.26	0.00028	25.78	0.37	0.21

alpha	alpha'	alpha''
1.94	3.46	1.62
7.99	6.74	2.05
8.21	6.44	2.04
8.77	6.24	2.05
9.14	6.06	2.05
5.18	16.96	2.28
5.48	17.20	2.30
5.93	14.97	2.31
6.25	13.80	2.31
6.53	12.88	2.32
4.68	22.50	2.32
5.02	23.83	2.35
5.28	22.13	2.36
5.57	20.26	2.38
5.79	19.18	2.38

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x2a	x2b	x2s	p2a	p2b
0.79	0.93	0.79	2645.9	3107.54
	0.91	0.60	2645.9	3047.33
	0.90	0.59	2645.9	3020.57
	0.89	0.57	2645.9	2987.12
	0.88	0.56	2645.9	2957.01
	0.97	0.66	5508.6	6755.16
	0.97	0.65	5508.6	6748.20
	0.96	0.62	5508.6	6692.48
	0.96	0.61	5508.6	6650.70
	0.95	0.59	5508.6	6608.91
	0.98	0.66	7962.3	9844.67
	0.98	0.65	7962.3	9844.67
	0.98	0.64	7962.3	9814.47
	0.97	0.62	7962.3	9774.20
	0.97	0.61	7962.3	9744.01

xBf	x2f	xAf+xBf
0.35	0.86	0.41
0.40	0.76	0.53
0.46	0.75	0.61
0.47	0.73	0.64
0.50	0.72	0.70
0.75	0.81	0.92
0.83	0.81	1.03
0.90	0.79	1.13
0.97	0.78	1.24
1.03	0.77	1.34
1.14	0.82	1.38
1.21	0.81	1.48
1.27	0.81	1.58
1.33	0.80	1.67
1.40	0.79	1.78

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Vf (m ³ /s)	Vp 1(m ³ /s)	Vr2 (m ³ /s)	Vr1
0.00028	0.00022	0.00006	0.00006
0.00036	0.00023	0.00013	0.00013
0.00041	0.00023	0.00018	0.00018
0.00043	0.00023	0.00020	0.00020
0.00047	0.00024	0.00023	0.00023
0.00062	0.00056	0.00006	0.00006
0.00069	0.00057	0.00012	0.00012
0.00076	0.00058	0.00018	0.00018
0.00083	0.00059	0.00024	0.00024
0.00090	0.00060	0.00030	0.00030
0.00093	0.00087	0.00006	0.00006
0.00100	0.00088	0.00012	0.00012
0.00107	0.00088	0.00018	0.00018
0.00113	0.00089	0.00024	0.00024
0.00120	0.00090	0.00030	0.00030

Pt (atm)	O2 Recovery	N2 Recovery
2.401	0.328	0.192
2.401	0.496	0.298
2.401	0.453	0.350
2.401	0.459	0.356
2.401	0.452	0.370
7.163	0.208	0.027
7.163	0.196	0.049
7.163	0.193	0.066
7.163	0.186	0.081
7.163	0.181	0.092
11.245	0.134	0.012
11.245	0.131	0.022
11.245	0.128	0.031
11.245	0.126	0.038
11.245	0.124	0.045

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	Pipe (gmol/min)	KO ₂	KN ₂	Ratio KO ₂ /KN ₂
50.00	0.04	0.000109	0.000056	1.94
50.00	0.10	0.000337	0.000042	7.99
50.00	0.14	0.000345	0.000042	8.21
50.00	0.15	0.000362	0.000041	8.77
50.00	0.17	0.000381	0.000042	9.14
120.00	0.04	0.000252	0.000049	5.18
120.00	0.09	0.000267	0.000049	5.48
120.00	0.13	0.000283	0.000048	5.93
120.00	0.18	0.000295	0.000047	6.25
120.00	0.23	0.000304	0.000047	6.53
180.00	0.04	0.000240	0.000051	4.68
180.00	0.09	0.000253	0.000050	5.02
180.00	0.13	0.000263	0.000050	5.28
180.00	0.18	0.000273	0.000049	5.57
180.00	0.22	0.000281	0.000048	5.79

8. Appendix II: Error Analysis (10 pts)

Potential sources of error may exist within the equipment, such as insufficient purging of the streams when adjusting flow rates and pressure. Additionally, the O₂ sensors may not be functioning or calibrated properly. A pressure leak from the compressed air tank could lead to a decrease in pressure within the system. Furthermore, the flow meter might be defective and unable to provide the correct flow rate into the system. There is also a possibility that the flow meter is mis-calibrated, resulting in a mismatch between the controller and the valve sensitivity. Despite these potential sources of error, none impacted the capacity to produce accurate results. These errors are purely theoretical and do not pertain to the experiment conducted. To prevent these types of errors, it is essential to check the pressure hose leading into the system, calibrate the O₂ sensor prior to the experiment, and allow a calibration period of 10 minutes. Moreover, ensure that the flow meter is in alignment with the controller through calibration. Lastly, remember to purge the valves when adjusting the flow rates and pressure entering the system. Additionally, due to the limited amount of compressed air within the air cylinder, replicate data was not obtained. If replicated data were obtained, the compressed air

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cylinder would run out of pressure. Moreover, if replicated sets were obtained then error analysis would consist of standard deviation, RSD(%) , variance, and error. This would allow us to gauge the data in sets to check for accuracy.

9. Appendix III: Sample Calculations (10 pts)

Here is a sample calculation for a singular run for the parallel configuration. The first run will be used for calculations. The collected pressure is in PSIG, and the collected flow rates are in SLPM. To convert from PSIG to torr, the following equation was used:

$$Pressure(torr) = \left(\frac{Pressure(PSIG) + 14.7}{14.7} \right) * 760.$$

To convert SLPM to gmol/min, the following equation was used:

$$\dot{m} \left(\frac{gmol}{min} \right) = \frac{m \cdot (SLPM) * 1000}{22400}. \text{ This equation is valid for feed, tube, and shell flow.}$$

Using values from the first parallel run, the calculated are as follows:

$$Pressure(torr) = \left(\frac{51 + 14.7}{14.7} \right) * 760 = 3400 \text{ torr}$$
$$\dot{m}_p \left(\frac{gmol}{min} \right) = \frac{3.25 * 1000}{22400} = 0.14 \text{ gmol/mol. This can be referred to as G}$$
$$\dot{m}_R \left(\frac{gmol}{min} \right) = \frac{0.98 * 1000}{22400} = 0.044 \text{ gmol/min. This can be referred to as F}$$
$$\dot{m}_F \left(\frac{gmol}{min} \right) = 0.14 + 0.044 = 0.184 \text{ gmol/min}$$

Afterwards, we can find the mass fractions of O₂ and N₂ in various flows:

X_{1a} is the feed, which is air. This is known to be 20.9% O₂ and 79.1% N₂

X_{1s} and X_{1b} are obtained from O₂ sensors. The X_{2s} and X_{2b} are derived by subtracting 1 by X_{1s} and X_{1b} respectively.

Using the values obtained in lab, X_{2s} = 1 – 0.298 = 0.702 and X_{2b} = 1 – 0.091 = 0.909

Moving on to O₂ partial pressures, P_{1a}, P_{1b}, P_{1t}, and P_{1s}. These are calculated as follows:

$$P_{1a} = Pressure(torr) * 0.209 = 3400 * 0.209 = 711 \text{ Torr}$$

$$P_{1b} = Pressure(torr) * x_{1b} = 3400 * 0.091 = 309 \text{ Torr}$$

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$$P_{1t} = \frac{P_{1a} + P_{1b}}{2} = \frac{711 + 309}{2} = 510 \text{ Torr}$$

$$P_{1s} = x_{1s} * 760 = 0.298 * 760 = 226 \text{ Torr}$$

The process for N₂ is similar:

$$P_{2a} = \text{Pressure (torr)} * (1 - 0.209) = 3400 * 0.791 = 2690 \text{ Torr}$$

$$P_{2b} = \text{Pressure (torr)} * (1 - x_{1b}) = 3400 * 0.909 = 3090 \text{ Torr}$$

$$P_{2t} = \frac{P_{2a} + P_{2b}}{2} = \frac{2690 + 3090}{2} = 2890 \text{ Torr}$$

$$P_{2s} = (1 - x_{1s}) * 760 = 0.702 * 760 = 534 \text{ Torr}$$

The O₂ permeation rate, g₁, is calculated as such:

$$g_1 = G * x_{1s} = 0.14 * 0.298 = 0.042 \text{ gmol/min}$$

The N₂ permeation rate, g₂, is calculated in a similar fashion:

$$g_2 = G * x_{2s} = 0.14 * 0.702 = 0.0983 \text{ gmol/min}$$

The transport coefficient for O₂, which is known as KO₂, is calculated by:

$$KO_2 = \frac{g_1}{(P_{1t} - P_{1s})} = \frac{0.042}{(510 - 226)} = 1.5\text{E-}4 \text{ gmol O}_2 / (\text{min torr})$$

The process is very similar to KN₂:

$$KN_2 = \frac{g_2}{(P_{2t} - P_{2s})} = \frac{0.0983}{(2890 - 534)} = 4.33\text{E-}5 \text{ gmol N}_2 / (\text{min torr})$$

To solve separation efficiency is based on the different rates of permeation of the gas components, α, the following equation is used:

$$\alpha_{AB} = \frac{Q_A}{Q_B} = \frac{1.5\text{E-}4}{4.33\text{E-}5} = 3.53$$

To solve for the separation factor based on retentate composition, α', the following equation is used:

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$$\alpha'_{AB} = \frac{\frac{X_{Ap}}{X_{Bp}}}{\frac{X_{Ar}}{X_{Br}}} = \frac{\frac{0.298}{0.702}}{\frac{0.091}{0.909}} = 4.24$$

To solve the separation factor based on feed composition, α'' , we use:

$$x_{Af} = \frac{(x_{1s} * \dot{m}_F) + (x_{1b} * \dot{m}_F)}{(x_{1s} * \dot{m}_F) + (x_{1b} * \dot{m}_F) + (x_{2B} * \dot{m}_F) + (x_{2s} * \dot{m}_F)} = \frac{(0.298 * 0.184) + (0.091 * 0.184)}{(0.298 * 0.184) + (0.091 * 0.184) + (0.909 * 0.184) + (0.702 * 0.184)} = 0.19$$

$$x_{Bf} = \frac{(x_{2B} * \dot{m}_F) + (x_{2s} * \dot{m}_F)}{(x_{1s} * \dot{m}_F) + (x_{1b} * \dot{m}_F) + (x_{2B} * \dot{m}_F) + (x_{2s} * \dot{m}_F)} = \frac{(0.909 * 0.184) + (0.702 * 0.184)}{(0.298 * 0.184) + (0.091 * 0.184) + (0.909 * 0.184) + (0.702 * 0.184)} = 0.81$$

$$\alpha''_{AB} = \frac{\frac{X_{Ap}}{X_{Bp}}}{\frac{X_{Af}}{X_{Bf}}} = \frac{\frac{0.298}{0.702}}{\frac{0.1945}{0.8055}} = 1.76$$

To find the recoveries of O₂ and N₂, the following equations are used:

$$V_p \left(\frac{m^3}{min} \right) = \dot{m}_p (SLPM) * 0.0010002 = 3.25 * 0.0010002 = 0.00325 \frac{m^3}{min}$$

$$V_f \left(\frac{m^3}{min} \right) = \dot{m}_f (SLPM) * 0.0010002 = 4.23 * 0.0010002 = 0.00423 \frac{m^3}{min}$$

$$V_r \left(\frac{m^3}{min} \right) = \dot{m}_r (SLPM) * 0.0010002 = 0.98 * 0.0010002 = 0.00098 \frac{m^3}{min}$$

The following equations were provided in the manual:

$$O_2 recovery = \left(\frac{V_p C_{O_{2p}}}{V_f C_{O_{2f}}} \right)$$

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$$N_2 recovery = \left(\frac{V_p C_{N_{2r}}}{V_f C_{N_{2f}}} \right)$$

To calculate the concentrations, ideal gas is assumed:

$$C = \frac{P}{RT}$$

And;

$$C_{O_{2p}} = x_1 \frac{s * P_s}{RT}$$

$$C_{O_{2f}} = x_1 \frac{a * P_f}{RT}$$

Plugging this in we get;

$$O_2 recovery = \left(\frac{V_p * x_1 \frac{s * P_s}{RT}}{V_f * x_1 \frac{a * P_t}{RT}} \right)$$

Which then simplifies to:

$$O_2 recovery = \left(\frac{V_p * x_1 s * P_s}{V_f * x_1 a * P_t} \right)$$

This assumption can also be made to calculate N2 recovery.

$$N_2 recovery = \left(\frac{V_r * x_2 b * P_s}{V_f * x_2 a * P_t} \right)$$

Plugging in the values:

$$O_2 recovery = \left(\frac{0.00325 * 0.298 * \cancel{i}}{V_f * x_1 a * P_t} \right) \cancel{i}$$

10. Appendix IV: Individual Team Contributions

NAME: Adil Feratovic

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	Assisted in conduction of lab
Pre-Lab Editing	1	Made objective portion
Experimental Objective (Prelab)	1	Created the objective
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)	2	Formulated an adequate procedure
Project Safety Evaluation (prelab)	0	
Final Lab Editing	8	Made final corrections and touch ups. Also, made sure each section is completed.
Executive Summary (final lab)	1	Created the conclusion and objective portion.
Introduction (Final lab)	0	
Theory (Final Lab)	0	
Results (final lab)	0	
Discussion (final lab)	0	
References (final and/or prelab)	0	
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	2	Assisted in the analysis and its formulation
Sample Calculations (final Lab)	5	Completed and tabulated sample calculations
Power Point Presentation	2	Assisted in the collaboration of the team creation
Total	26	

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NAME: Hussein Rezk

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	Read and conducted an experiment
Pre-Lab Editing	3	Assisted in all parts
Experimental Objective (Prelab)	2	Assisted in objective creation
Apparatus (prelab)	1	Assisted in the assembly of the apparatus
Materials & Supplies (prelab)	1	Created a list of materials and supplies
Procedure (prelab)	2	Helped create a procedure for the experiment report
Project Safety Evaluation (prelab)	1	Made safety guidelines
Final Lab Editing	3	Final grammar touch-ups
Executive Summary (final lab)	2	Assisted in the formation of the summary
Introduction (Final lab)	0	
Theory (Final Lab)	3	Found valuable literature and theory to perform an analytical comparison
Results (final lab)	0	
Discussion (final lab)	4	Analyzed correlations, theory linearity, and the correctness of experiment results
References (final and/or prelab)	1	Tabulated references
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	1	Conducted analysis on data
Sample Calculations (final Lab)	0	
Power Point Presentation	2	Created in collaboration with group
Total	30	

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NAME: Aaron Casas

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	Followed lab procedure
Pre-Lab Editing	3	Worked on the prelab
Experimental Objective (Prelab)	1	
Apparatus (prelab)	.5	Picture descriptions
Materials & Supplies (prelab)	.5	Material and supplies description
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	.5	Took in consideration dangers of pressurized gas
Final Lab Editing	5	Reviewed calculations to make sure numbers made sense
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	0	
Results (final lab)	4	Provided brie descriptions of graphs and tables.
Discussion (final lab)	0	
References (final and/or prelab)	0	
Data Tabulation/Graphs (final)	2	Format graph and tables to report relevant information
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	2	Calculated values required from the lab report
Power Point Presentation	0	
Total	22.5	

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NAME: Anna Cai

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	2	Collected data and helped conduct experiment
Pre-Lab Editing	2	Edited prelab
Experimental Objective (Prelab)	0	
Apparatus (prelab)	1	Took photos and wrote description of apparatus
Materials & Supplies (prelab)	1	
Procedure (prelab)	1	Helped write procedure
Project Safety Evaluation (prelab)	0	
Final Lab Editing	2	Reviewed grammar and organized document
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	1	
Results (final lab)	5	Completed calculations
Discussion (final lab)	1	Analyzed results
References (final and/or prelab)	0.5	
Data Tabulation/Graphs (final)	5	Created graphs and tables based on results
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	3	Completed sample calculations
Power Point Presentation	0	
Total	24.5	